

Medical Insight Explorer Agent

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Bilingual conversational healthcare analytics with deterministic insights.

LangGraph + Gradio + Hugging Face

← → ↻ huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent 🔍 ☆ 📄 | A ⋮

 Spaces |  Artur-Melnyk/Medical-Insight-Explorer-Agent 📄  like 0  Agents  Running  Logs

 App  Files  Community  Settings ⋮ 

Medical Insight Explorer Agent

Conversational analytics over cleaned Medicare healthcare claims data.
Konversationelle Analyse bereinigter Medicare-Healthcare-Claims-Daten.

Deterministic analytics first. Grounded explanations and optional visual outputs.

Language / Sprache

English ▼

Ask a healthcare analytics question / Frage zur Healthcare-Analyse

Show top inpatient providers

Run analysis / Analyse ausführen Clear / Zurücksetzen

≡ English examples

≡ Examples

Show me the shape of all tables Give me an inpatient summary Give me an outpatient summary What is the average beneficiary age? Show top inpatient providers

Show top outpatient providers Show inpatient claims by state What is the diabetes cost summary? Show reimbursement distribution

Business Problem

- Healthcare claims data is large and difficult to explore
 - Analysts need grounded and deterministic outputs
- LLM systems in healthcare require explainability and safety
 - Goal: fast analytics without clinical advice

Project Ecosystem

Healthcare Data Cleaning Repo



Validated Parquet Data



Medical Insight Explorer Agent



Gradio + LangGraph



Hugging Face Deployment

Architecture



Technical Design Choices

Engineering decisions used to improve reliability, safety, and deployment stability

Design Decision	Why it was implemented
Compute First → Explain Second	Deterministic pandas analytics run before AI explanations to prevent fabricated healthcare statistics
Controlled Analytics Routing	User questions are routed only to predefined functions; the LLM never receives direct DataFrame access
Sample Data Deployment Strategy	Lightweight sample datasets were used to satisfy Hugging Face memory and runtime constraints
Localized Bilingual UX	English and German interaction improves accessibility and recruiter usability
Conditional Visualization Routing	Plotly charts are generated only for supported analytical routes to avoid misleading outputs

Design principle: Safe computation → controlled reasoning → explainable output

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Conversational analytics over cleaned Medicare healthcare claims data.

This app uses deterministic pandas-based analytics functions first, then returns grounded explanations and optional visual outputs.

Ask a healthcare analytics question

Show top inpatient providers

Run analysis

Clear

Examples

Show me the shape of all tables

Give me an inpatient summary

Give me an outpatient summary

What is the average beneficiary age?

Show top inpatient providers

Show top outpatient providers

Show inpatient claims by state

What is the diabetes cost summary?

Show reimbursement distribution

Agent response

Route selected: top_inpatient_providers

Explanation:

Route selected: top_inpatient_providers. The result was computed using deterministic pandas-based analytics. Computed output: [{"Provider": "PRV52019", "claim_count": 516}, {"Provider": "PRV55462", "claim_count": 386}, {"Provider": "PRV54367", "claim_count": 322}, {"Provider": "PRV53706", "claim_count": 282}, {"Provider": "PRV55209", "claim_count": 275}, {"Provider": "PRV56560", "claim_count": 248}, {"Provider": "PRV54742", "claim_count": 231}, {"Provider": "PRV55230", "claim_count": 225}, {"Provider": "PRV52340", "claim_count": 224}, {"Provider": "PRV51501", "claim_count": 223}]

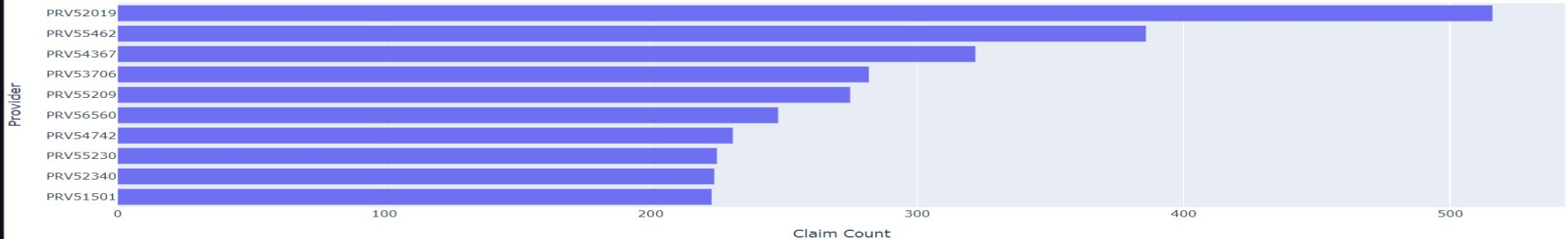
Computed result:

Provider	claim_count
PRV52019	516
PRV55462	386
PRV54367	322
PRV53706	282

PRV52019	516
PRV55462	386
PRV54367	322
PRV53706	282
PRV55209	275
PRV56560	248
PRV54742	231
PRV55230	225
PRV52340	224
PRV51501	223

Visualization output

Top 10 Inpatient Providers by Claim Count

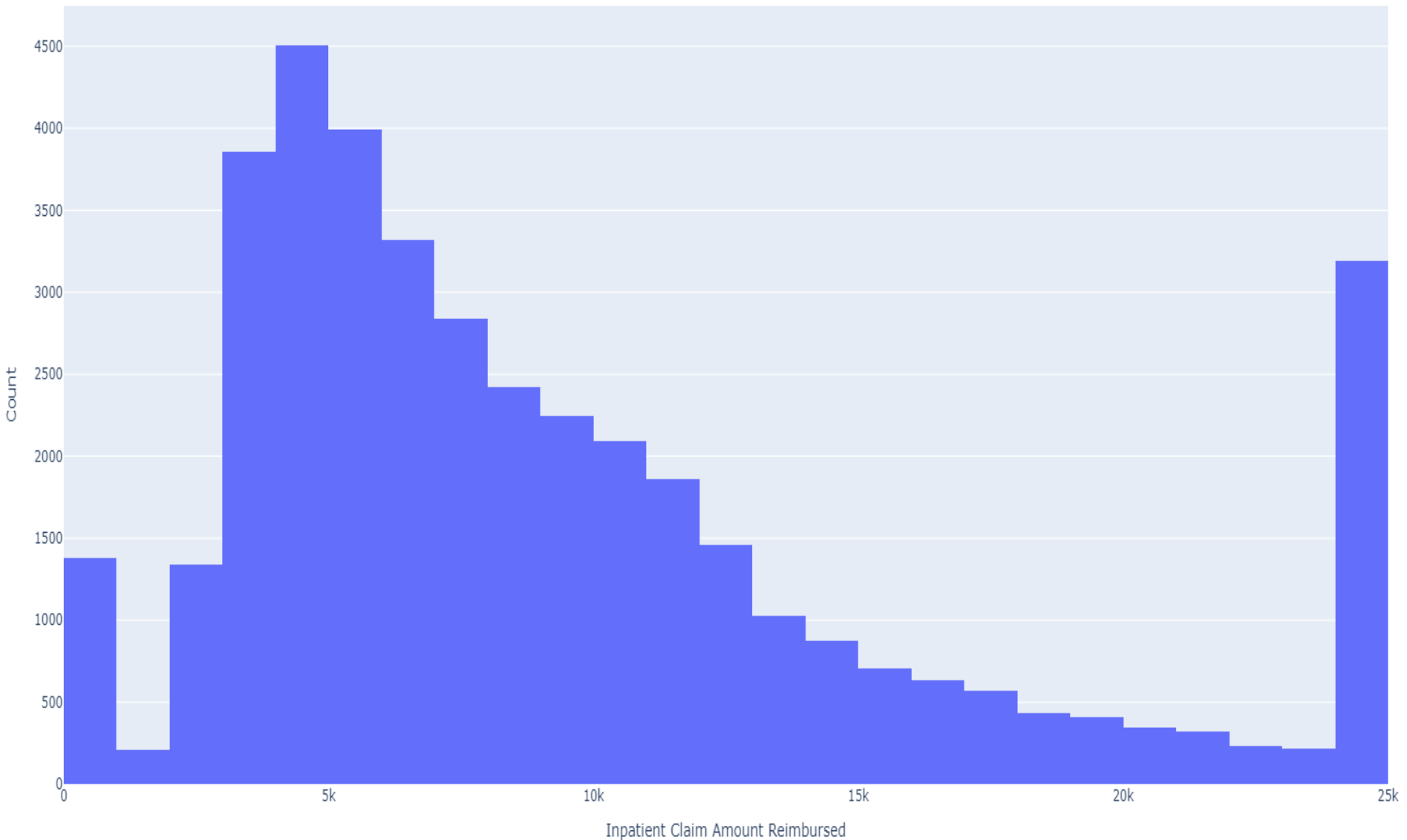


Key Features

- ✓ English/German UX
 - ✓ Plotly charts
- ✓ Deterministic analytics
- ✓ LangGraph orchestration
- ✓ Safe compute-first architecture

English Demo

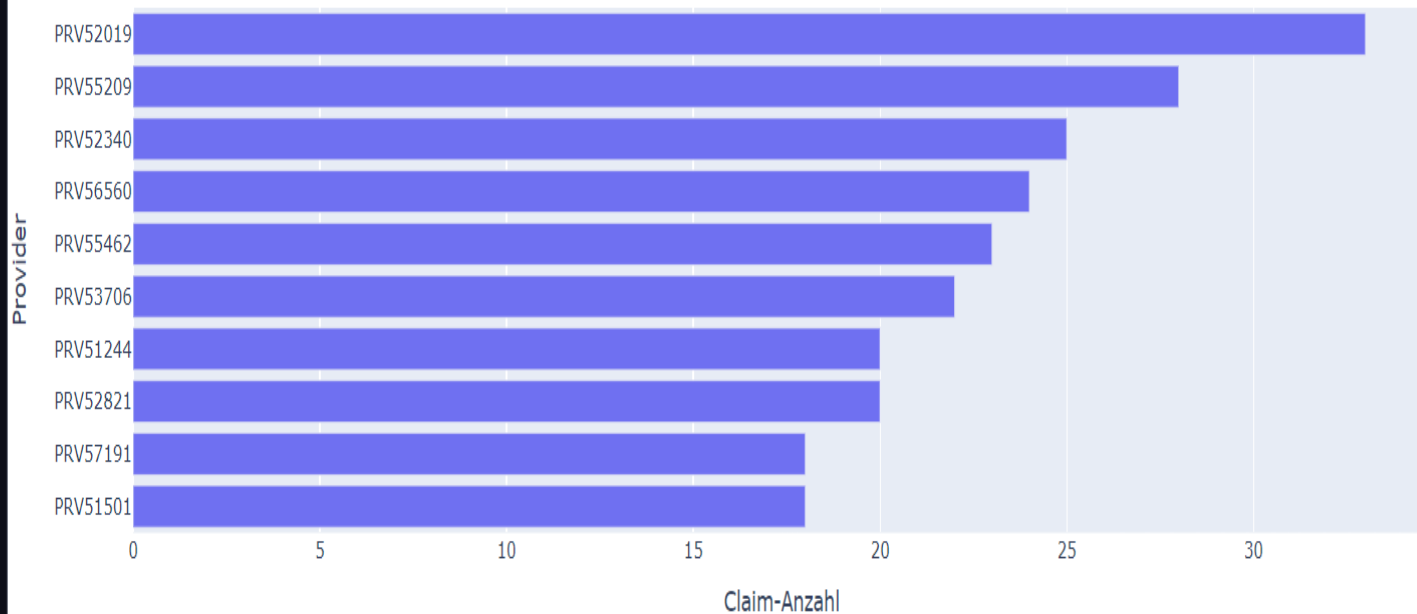
Distribution of Inpatient Claim Reimbursement



German Demo

Visualization output / Visualisierung

Top 10 stationäre Provider nach Claim-Anzahl

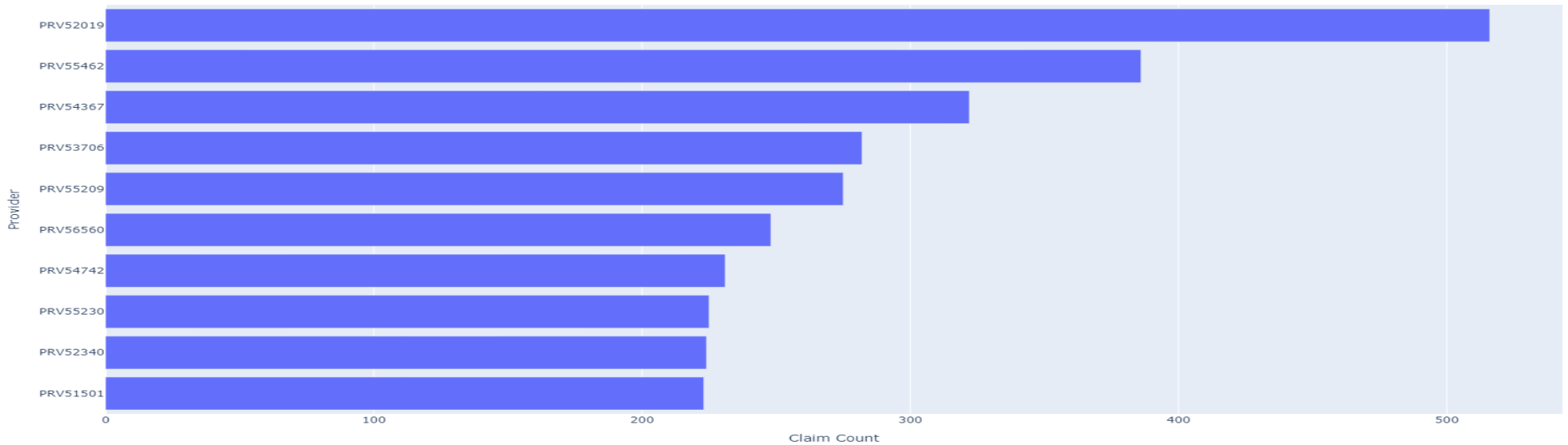


Supported languages: English | Deutsch

Deterministic analytics • Claims data only • No diagnosis

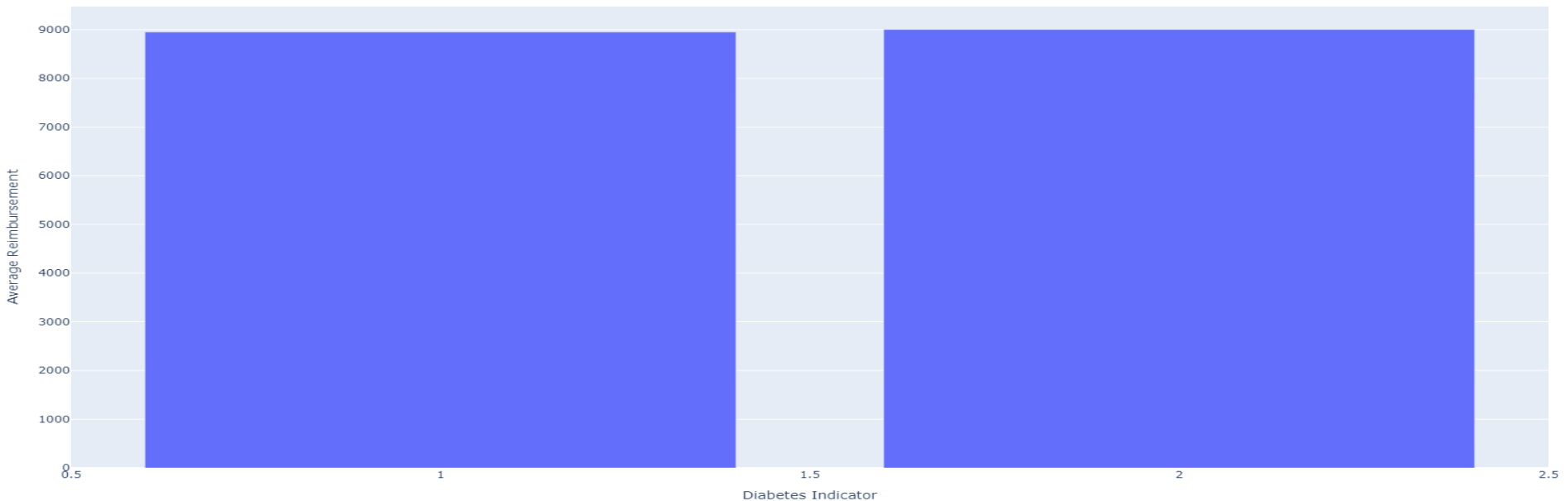
Analytics Outputs

Top 10 Inpatient Providers by Claim Count



Insight: Provider activity is concentrated among a small subset of institutions.

Average Inpatient Reimbursement by Diabetes Indicator



Insight: Reimbursements are right-skewed with a concentration around lower claim amounts.

Safety & Responsible AI

- ✓ Claims analytics only
- ✓ No diagnosis
- ✓ No treatment recommendations
- ✓ Explainable deterministic calculations
- ✓ Safer healthcare AI interaction

Live Demo & Portfolio

Key Takeaways

- ✓ Safe healthcare analytics
- ✓ Explainable AI workflow
- ✓ Production deployment
- ✓ Bilingual recruiter-ready UX

Hugging Face:

<https://huggingface.co/spaces/Artur-Melnyk/Medical-Insight-Explorer-Agent>

GitHub:

<https://github.com/ArturMelnyk-analyst/Medical-Insight-Explorer-Agent>

Built using: Python • Pandas • Plotly • Gradio • LangGraph • Hugging Face